

BUSINESS CASE STUDY- TARGET

- I. Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:

- i. Data type of all columns in the “customers” table.

Result

	customers	 Query	Open in ▾	 Share ▾
Schema	Details	Preview	Table explorer	Preview
Filter Filter tree				
☐	Field name	Type	Mode	
☐	customer_id	STRING	NULLABLE	
☐	customer_unique_id	STRING	NULLABLE	
☐	customer_zip_code_prefix	INTEGER	NULLABLE	
☐	customer_city	STRING	NULLABLE	
☐	customer_state	STRING	NULLABLE	

Insights

- There are Four (4) String Types and One (1) Integer Type.
- String Type: customer_id, customer_unique_id, customer_city and customer_state.
- Integer Type: customer_zip_code_prefix.

- ii. Get the time range between which the orders were placed.

Query

SELECT

MIN(order_purchase_timestamp) as first_order,

MAX(order_purchase_timestamp) as last_order

FROM target.orders;

Result

Query results		
Job information	Results	Visualisation JSON
Row	first_order ▾	last_order ▾
1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC

Insights:

- The First Order was made on 04th September 2016 at 21 Hours 15 Minutes 19 Second UTC and The Last Order was made on 17th October 2018 at 17 Hours 30 Minutes 18 Second UTC.
- The time period between First order and the last order is 2years 1month and 13days.

iii. Count the Cities & States of customers who ordered during the given period.

Query

SELECT

```
EXTRACT(Year from o.order_purchase_timestamp) AS Year,  
COUNT(DISTINCT tc.customer_city) AS City_cnt,  
COUNT(DISTINCT tc.customer_state) as State_cnt
```

```
FROM `target.customers`tc JOIN `target.orders`o
```

```
ON tc.customer_id = o.customer_id
```

```
GROUP BY Year
```

```
ORDER BY Year;
```

Result

Query results

Job information		Results	Visualisation	JSON
Row	Year ▼	City_cnt ▼	State_cnt ▼	
1	2016	179	21	
2	2017	3290	27	
3	2018	3277	27	

Insights:

- In **2016**, orders are placed from **179 cities across 21 states**, showing a small but focused coverage.
- In **2017**, the reach grew rapidly to **3,290 cities across 27 states**, which shows there is a major geographic expansion.
- In **2018**, the reach slightly decreases to **3,277 cities across the same 27 states**, which shows there is a saturation in market presence.

II. In-depth Exploration:

- i. Is there a growing trend in the no. of orders placed over the past years?

Query

```
SELECT
EXTRACT(Year FROM order_purchase_timestamp) AS Year,
COUNT(order_id) as Total_order
FROM `target.orders`
GROUP BY Year
ORDER BY Year;
```

Result

Query results			
Job information		Results	Visualise
Row	Year	Total_order	
1	2016	329	
2	2017	45101	
3	2018	54011	

Insights:

- In **2016**, the number of orders placed was very few (only **329** orders).
- In **2017**, the number of orders jumped massively to **45,101** showing the growth of the users.
- In **2018**, orders further increased to **54,011** which shows the continued expansion and an increasing customer base.

So here, the above analysis clearly shows the growing trend in the number of orders placed over the years.

- ii. Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

Query

```
SELECT
EXTRACT(Year FROM order_purchase_timestamp) AS Year,
COUNT(order_id) as Total_order,
CASE WHEN EXTRACT(Month FROM order_purchase_timestamp) in (1,2,3) THEN "Winter"
      WHEN EXTRACT(Month FROM order_purchase_timestamp) in (4,5,6) THEN "Spring"
      WHEN EXTRACT(Month FROM order_purchase_timestamp) in (7,8,9) THEN "Summer"
      ELSE "Autumn" End As Monthly_Seasonality
FROM `target.orders`
GROUP BY Year, Monthly_Seasonality
ORDER BY Year;
```

Result

Query results

Job information		Results	Visualisation	JSON
Row	Year ▼	Total_order ▼	Monthly_Seasonality ▼	
1	2016	328	Autumn	
2	2016	1	Winter	
3	2017	8253	Winter	
4	2017	8786	Spring	
5	2017	11602	Summer	
6	2017	16460	Autumn	
7	2018	13997	Winter	
8	2018	21023	Spring	
9	2018	18971	Summer	
10	2018	20	Autumn	

Insights:

- In **2016**, mostly the orders were placed in **Autumn season** i.e., **328** and only **One (1)** order were placed in **Winter**. There was no order placed in **Spring and Summer season**.
- In **2017**, **Autumn (16,460 orders)** had the highest number of orders. While, **Summer** had the good number of orders (**11,602**) but orders in **Spring and Summer** is quiet low compare to **Autumn and Winter (Orders are in between 8,200 to 8,800)**.
- In **2018**, highest number of orders were made in **Spring season (21,023)** and lowest number of orders was made in **Autumn** which is only 20 orders. **Summer and Winter** had the good number of orders which is **18,971 and 13,997** respectively.

This shows that Customer orders show a strong year-on-year growth with clear seasonal peaks, especially during **Autumn and Spring**, while **Winter** showed lower but consistent demand. This indicates a strong overall trajectory with clear seasonal spikes in customer activity.

- iii. During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)
 - 0-6 hrs : Dawn
 - 7-12 hrs: Mornings
 - 13-18 hrs: Afternoon
 - 19-23 hrs: Night

Query

```
SELECT
COUNT(order_id) AS Total_order,
CASE
  WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 0 AND 6 THEN "Dawn"
  WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 7 AND 12 THEN
"Mornings"
  WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 13 AND 18 THEN
"Afternoon"
  ELSE "Night" END AS Time_of_day
FROM `target.customers`tc Join `target.orders` o
on tc.customer_id = o.customer_id
GROUP BY Time_of_day
ORDER BY Total_order desc;
```

Result

Query results

Job information		Results	Visualisation
Row	Total_order	Time_of_day	
1	38135	Afternoon	
2	28331	Night	
3	27733	Mornings	
4	5242	Dawn	

Insights

- During **Afternoon(13-18hrs)** there is a highest number of orders (**38,135**), making it the peak period for customer activity.
- Both **Night (28331)** and **Mornings (27733)** the number of orders is quite impressive but is less than the order made by the customers during **Afternoon**.
- The lowest number of orders made by the Brazilian customers is observed during **Dawn**. The number of orders made at this period is only **5242** showing that less customers are engaged during this period.

This shows that customers engagement is strongest in the **Afternoon** followed **Night and Mornings**, while **Dawn** sees the limited activity. This shows that Marketing and operational strategies should prioritize the **Afternoon and Night slots** to maximize impact.

III. Evolution of E-commerce orders in the Brazil region:

- i. Get the month-on-month no. of orders placed in each state.

Query

SELECT

```
FORMAT_DATE("%Y-%m",o.order_purchase_timestamp) AS Year_Month,  
c.customer_state,  
COUNT(DISTINCT o.order_id) AS Total_Customer
```

```
FROM `target.customers` c JOIN `target.orders` o
```

```
ON c.customer_id = o.customer_id
```

```
GROUP BY Year_Month, c.customer_state
```

```
ORDER BY Year_Month, c.customer_state;
```

Result

Query results

Job information		Results	Visualisation	JSON	Execution details
Row	Year_Month	customer_state	Total_Customer		
1	2016-09	RR	1		
2	2016-09	RS	1		
3	2016-09	SP	2		
4	2016-10	AL	2		
5	2016-10	BA	4		
6	2016-10	CE	8		
7	2016-10	DF	6		
8	2016-10	ES	4		
9	2016-10	GO	9		
10	2016-10	MA	4		

Insights:

- **São Paulo (SP-41,746), Rio de Janeiro (RJ-12,852), Minas Gerais (MG -11,635), Rio Grande do Sul (RS-5,466), Paraná (PR-5,045) and Santa Catarina (SC-3,637)** are the top most contributing states in terms of customers. Together they contribute nearly around 75% of total customers.
- States with much smaller presence include **Roraima (RR), Amapá (AP), Acre (AC)**, each contributing very few customers.
- There is upward trend from late **2016** to mid-**2018**.
- **Nov 2017(7,544), Jan 2018(7,269), Mar 2018(7,211), Apr 2018(6,939) and May 2018(6,873)** shows the **TOP CUSTOMER MONTHS**.
- The business expanded rapidly in **2017–2018**, with **São Paulo (SP)** as the dominant market.
- Seasonal peaks suggest possible **holiday**

ii. How are the customers distributed across all the states?

Query

```
SELECT
    customer_state,
    COUNT(DISTINCT customer_id) AS Total_customer
FROM `target.customers`
GROUP BY customer_state
ORDER BY Total_customer desc;
```

Result

Query results

Job information			Results	Visualisation	J
Row	customer_state	Total_customer			
1	SP	41746			
2	RJ	12852			
3	MG	11635			
4	RS	5466			
5	PR	5045			
6	SC	3637			
7	BA	3380			
8	DF	2140			
9	ES	2033			
10	GO	2020			

Insights

- The majority of customers are from **São Paulo (SP)** with **41,756 customers**, which is far higher than any other state.
- **São Paulo (SP-41,746)**, **Rio de Janeiro (RJ-12,852)**, **Minas Gerais (MG-11,635)**, **Rio Grande do Sul (RS-5,466)**, **Paraná (PR-5,045)** and **Santa Catarina (SC-3,637)** are the top most contributing states in terms of customers. Together they contribute nearly around 75% of total customers.
- States such as **Pará (PA-975)**, **Ceará (CE-1,336)** and **Pernambuco (PE-1,652)** shows a small number of customers.
- States like **Acre (AC-81)**, **Amapá (AP-68)** and **Roraima (RR-46)** has the lowest customers counts, shows the limit market penetration.
- The difference between top state **São Paulo (SP -41,756)** and the lowest **Roraima (RR-46)** shows that there is a very uneven customers distribution across **Brazil**.

IV. Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.

- i. Get the % increase in the cost of orders from year 2017 to 2018 (*include months between Jan to Aug only*).

You can use the “payment_value” column in the payments table to get the cost of orders.

Query

```
WITH payments_with_year_month AS (  
  SELECT  
    p.order_id,  
    p.payment_value,  
    EXTRACT(YEAR FROM o.order_purchase_timestamp) AS order_year,  
    EXTRACT(MONTH FROM o.order_purchase_timestamp) AS order_month  
FROM target.payments p JOIN target.orders o  
  ON p.order_id = o.order_id),  
filtered_payments AS (  
  SELECT  
    order_year,  
    ROUND(SUM(payment_value),2) AS total_payment  
FROM payments_with_year_month  
WHERE order_month BETWEEN 1 AND 8  
AND order_year IN (2017, 2018)  
GROUP BY order_year)  
SELECT  
  SUM(CASE WHEN order_year = 2017 THEN total_payment END) AS total_2017,  
  SUM(CASE WHEN order_year = 2018 THEN total_payment END) AS total_2018,  
  ROUND(((SUM(CASE WHEN order_year = 2018 THEN total_payment END) - SUM(CASE WHEN  
order_year = 2017 THEN total_payment END)) / SUM(CASE WHEN order_year = 2017 THEN  
total_payment END)) * 100, 2) AS percentage_increase  
FROM filtered_payments;
```

Result

Query results

Job information		Results	Visualisation	JSON
Row	total_2017	total_2018	percentage_increase	
1	3669022.12	8694733.84	136.98	

Insights

- Between **January and August**, the total cost of orders increased from **3669022.12(3.6M)** in **2017** to **8694733.84(8.6M)** in **2018** which shows there is an increase of **136.98%** over the year.
- This shows that the revenue increases a **more than a double in a year for the same time period**.

ii. Calculate the Total & Average value of order price for each state.

Query

SELECT

```
c.customer_state,  
ROUND(SUM(price),2) AS Total_price,  
ROUND(AVG(price),2) AS Avg_price  
FROM `target.customers` c JOIN `target.orders` o  
ON c.customer_id = o.customer_id JOIN `target.order_items` ot  
ON o.order_id = ot.order_id  
GROUP BY c.customer_state  
ORDER BY Total_price desc;
```

Result

Query results

Job information		Results	Visualisation	JSON	Execution
Row	customer_state	Total_price	Avg_price		
1	SP	5202955.05	109.65		
2	RJ	1824092.67	125.12		
3	MG	1585308.03	120.75		
4	RS	750304.02	120.34		
5	PR	683083.76	119.0		
6	SC	520553.34	124.65		
7	BA	511349.99	134.6		
8	DF	302603.94	125.77		
9	GO	294591.95	126.27		
10	ES	275037.31	121.91		

Insights:

- **São Paulo (SP)** has the highest **total order value (52022955.05 ~ 5M)** compare to all other states in Brazil. Whereas, **Rio de Janeiro (RJ) (1824092.67 ~ 1.8M)** and **Minas Gerais (MG) (1585308.03 ~ 1.5M)** also contributes significantly. These three states of Brazil together contribute majority of revenue.
- **Paraíba (PB)** has the highest **average order value (191.48)**, despite a much smaller total compared to **São Paulo (SP)**. States like **Alagoas (AL-180.89)** and **Piauí (PI-160.36)** also show high per-order spending.
- **Roraima (RR)** has the lowest total order value (**7829.43**) and also a relatively low average order value (**150.57**). This indicates limited customer base and order activity.

iii. Calculate the Total & Average value of order freight for each state.

Query

SELECT

c.customer_state,

ROUND(SUM(freight_value),2) AS Total_freight,

ROUND(AVG(freight_value),2) AS Avg_freight

FROM `target.customers` c JOIN `target.orders` o

ON c.customer_id = o.customer_id JOIN `target.order\_items` ot

ON o.order_id = ot.order_id

GROUP BY c.customer_state

ORDER BY Total_freight desc;

Result

Query results

Job information		Results	Visualisation	JSON	Executi
Row	customer_state	Total_freight	Avg_freight		
1	SP	718723.07	15.15		
2	RJ	305589.31	20.96		
3	MG	270853.46	20.63		
4	RS	135522.74	21.74		
5	PR	117851.68	20.53		
6	BA	100156.68	26.36		
7	SC	89660.26	21.47		
8	PE	59449.66	32.92		
9	GO	53114.98	22.77		
10	DF	50625.5	21.04		

Insights

- **SP (São Paulo)** has the highest freight value (**718723.07 ~ 718K**) compare to any other states in Brazil. This shows that **SP** dominates in order volume, driving up logistic costs,
- States like **RJ (Rio de Janeiro)**, **MG (Minas Gerais)**, **RS (Rio Grande do Sul)** and **PR(Paraná)** also contribute significantly, each cross more than 100K in total freight value.
- The highest average freight costs are observed in **less populated states** like **Roraima** (RR -42.98), **Paraíba** (PB-42.72) and **Rondônia** (RO-41.07). This indicates **higher shipping costs per order in less accessible states**, probably due to logistics challenges.
- **SP (São Paulo)** having the highest total freight but having one of the **lowest average freight values (15.15)** which shows that higher order density reduces per order shipping cost.

V. Analysis based on sales, freight and delivery time.

- i. Find the no. of days taken to deliver each order from the order's purchase date as delivery time. Also, calculate the difference (in days) between the estimated & actual delivery date of an order.

Do this in a single query.

You can calculate the delivery time and the difference between the estimated & actual delivery date using the given formula:

- $\text{time_to_deliver} = \text{order_delivered_customer_date} - \text{order_purchase_timestamp}$.
- $\text{diff_estimated_delivery} = \text{order_estimated_delivery_date} - \text{order_delivered_customer_date}$

Query

SELECT

order_id,

DATE_DIFF(order_delivered_customer_date, order_purchase_timestamp, DAY) AS

Time_to_deliver,

DATE_DIFF(order_estimated_delivery_date, order_delivered_customer_date, DAY) AS

Diff_estimated_delivery

FROM `target.orders`

WHERE order_delivered_customer_date IS NOT NULL

AND order_estimated_delivery_date IS NOT NULL

ORDER BY Time_to_deliver desc;

Result

Query results

Job information		Results	Visualisation	JSON	Executio
Row	order_id	Time_to_deliver	Diff_estimated_d...		
1	ca07593549f1816d26a572e06...	209	-181		
2	1b3190b2dfa9d789e1f14c05b6...	208	-188		
3	440d0d17af552815d15a9e41a...	195	-165		
4	285ab9426d6982034523a855f...	194	-166		
5	0f4519c5f1c541ddec9f21b3bd...	194	-161		
6	2fb597c2f772eca01b1f5c561bf...	194	-155		
7	47b40429ed8cce3aee9199792...	191	-175		
8	2fe324feb907e3ea3f2aa96508...	189	-167		
9	2d7561026d542c8dbd8f0daea...	188	-159		
10	c27815f7e3dd0b926b5855262...	187	-162		

Insights

- Most of the deliveries are in between **10 to 12 days**.
- The fastest delivery is on the same day delivery (**0 days**) and the slowest is (**209 days**).
- On average, orders are delivered nearly **11 days as promised**.
- The earliest delivery is **-188 days** which is way earlier than the estimated date and the maximum time taken to deliver the item is **+146 days**.

Deliveries are fairly quick on an average of **12days** but the **estimated delivery promises are too optimistic** since the customers usually has to wait an extra of 11 days.

- Find out the top 5 states with the highest & lowest average freight value.

Query

```
(SELECT c.customer_state,
       ROUND(AVG(freight_value),2) AS Avg_freight_value
FROM `target.customers` c JOIN `target.orders` o
ON c.customer_id = o.customer_id JOIN `target.order_items` ot
ON o.order_id = ot.order_id
GROUP BY c.customer_state
ORDER BY Avg_freight_value DESC
```

LIMIT 5)

UNION ALL

```
(SELECT c.customer_state,
       ROUND(AVG(freight_value),2) AS Avg_freight_value
FROM `target.customers`c JOIN `target.orders`o
ON c.customer_id = o.customer_id JOIN `target.order_items`ot
ON o.order_id = ot.order_id
GROUP BY c.customer_state
ORDER BY Avg_freight_value ASC
LIMIT 5);
```

Result

Query results		
Job information		
Results		
Visualisation		
Row	customer_state	Avg_freight_value
1	RR	42.98
2	PB	42.72
3	RO	41.07
4	AC	40.07
5	PI	39.15
6	SP	15.15
7	PR	20.53
8	MG	20.63
9	RJ	20.96
10	DF	21.04

Insights

- The top 5 states for highest average freight values are **Roraima** (RR-42.98), **Paraíba** (PB-42.72), **Rondônia** (RO-41.07), **Acre** (AC-40.07) and **Piauí** (PI-39.1). This shows that deliveries to the states like **RR, PB, RO, AC and PI** are costlier on average. These might be because they might be far from Brazil's main commerce hubs and may require longer logistics routes, raising costs.
- The lowest 5 states for lowest average freight values are **São Paulo** (SP-15.15), **Paraná** (PR-20.53), **Minas Gerais** (MG-20.63), **Rio de Janeiro** (RJ-20.96) and **Distrito Federal** (DF-21.04). Deliveries to **SP, RJ, MG, PR, and DF** are cheaper on average. These might be because these states might have better logistics infrastructure, shorter distances from warehouses, and higher delivery density.

Serving **SP, RJ, MG, PR, DF** is more cost-efficient due to concentrated demand and better transport networks than **RR, PB, RO, AC and PI**.

iii. Find out the top 5 states with the highest & lowest average delivery time.

Query

```
WITH delivery_times AS (  
    SELECT  
        c.customer_state,  
        DATE_DIFF(o.order_delivered_customer_date, o.order_purchase_timestamp, DAY) AS  
delivery_time  
    FROM `target.customers` c JOIN `target.orders` o  
        ON c.customer_id = o.customer_id  
    WHERE o.order_delivered_customer_date IS NOT NULL),  
avg_delivery AS (  
    SELECT  
        customer_state,  
        ROUND(AVG(delivery_time), 2) AS avg_delivery_time  
    FROM delivery_times  
    GROUP BY customer_state)  
(  
    SELECT customer_state, avg_delivery_time  
    FROM avg_delivery  
    ORDER BY avg_delivery_time DESC  
    LIMIT 5  
)  
  
UNION ALL  
  
(  
    SELECT customer_state, avg_delivery_time  
    FROM avg_delivery  
    ORDER BY avg_delivery_time ASC  
    LIMIT 5);
```

Result

Query results		
Job information Results Visualisation		
Row	customer_state	avg_delivery_time
1	RR	28.98
2	AP	26.73
3	AM	25.99
4	AL	24.04
5	PA	23.32
6	SP	8.3
7	PR	11.53
8	MG	11.54
9	DF	12.51
10	SC	14.48

Insights

- **Roraima** (RR-28.98 days), **Amapá** (AP-26.73 days), and **Amazonas** (AM-25.99 days) show the **longest delivery times**. These delays might be because of geographical distance from distribution hubs or logistical challenges. Customers in these states may experience frustration due to consistently high wait times.
- **São Paulo** (SP-8.3 days), **Paraná** (PR-11.53 days), and **Minas Gerais** (MG-11.54 days) have the **fastest delivery times**. These states might be well-connected with strong logistics infrastructure. Faster delivery here can boost customer satisfaction and improve the purchase rate.

There is a **huge gap** between the **fastest (8.3 days in SP)** and slowest (**28.9 days in RR**) delivery times almost **3.5x difference**. This highlights **logistical inequality**-customers who are far from the main hubs face longer waits, while those in major urban centers enjoy quicker service. Businesses may need to **optimize supply chain routes** or consider **regional warehouses** to reduce delivery delays in states with higher averages.

- iv. Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.

You can use the difference between the averages of actual & estimated delivery date to figure out how fast the delivery was for each state.

Query

```
WITH DATES AS (SELECT c.customer_state,
    DATE(o.order_delivered_customer_date) AS Actual_date,
    DATE(o.order_estimated_delivery_date) AS Estimated_delivery_date
FROM `target.customers` c JOIN `target.orders` o
ON c.customer_id = o.customer_id
WHERE o.order_delivered_customer_date IS NOT NULL
AND o.order_estimated_delivery_date IS NOT NULL
)
SELECT customer_state,
ROUND(AVG(DATE_DIFF(Estimated_delivery_date, Actual_date, DAY)),2) AS Fast_delivery
FROM DATES
GROUP BY customer_state
ORDER BY Fast_delivery DESC
LIMIT 5;
```

Result

Query results

Job information		Results	Visualisation
Row	customer_state	Fast_delivery	
1	AC	20.73	
2	RO	20.1	
3	AP	19.69	
4	AM	19.57	
5	RR	17.29	

Insights

- **Acre (AC)** shows the **fastest delivery performance**, with orders being delivered on average **20.73 days earlier** than the estimated delivery date. **Rondônia (RO)** is the second fastest, with an average early delivery of **20.1 days**, showing logistics efficiency in that region. **Amapá (AP)** also performs strongly, averaging **19.69 days**, indicating consistent delivery reliability. **Amazonas (AM)** comes next, with customers receiving their orders **19.57 days earlier** than expected. **Roraima (RR)** completes the top 5, with an average of **17.29 days faster delivery**.

- Customers in these states are experiencing significantly faster deliveries, which could improve customer satisfaction and loyalty.
- In reality, the **actual logistics performance is far better**, creating a big gap between estimated and actual delivery.

VI. Analysis based on the payments:

- Find the month-on-month no. of orders placed using different payment types.

Query

```
SELECT
FORMAT_DATE("%Y-%m", o.order_purchase_timestamp) AS Month,
p.payment_type,
COUNT(DISTINCT o.order_id) AS Total_order
FROM `target.orders` o JOIN `target.payments` p
ON o.order_id = p.order_id
WHERE o.order_purchase_timestamp IS NOT NULL
GROUP BY Month, p.payment_type
ORDER BY Month, Total_order DESC;
```

Result

Query results

Job information		Results	Visualisation	JSON	Execution details
Row	Month	payment_type	Total_order		
1	2016-09	credit_card	3		
2	2016-10	credit_card	253		
3	2016-10	UPI	63		
4	2016-10	voucher	11		
5	2016-10	debit_card	2		
6	2016-12	credit_card	1		
7	2017-01	credit_card	582		
8	2017-01	UPI	197		
9	2017-01	voucher	33		
10	2017-01	debit_card	9		

Insights

- Through **Credit Card**, there are nearly **76,505 orders**, which is the most preferred payment method. This holds the majority share of total transactions.

- **UPI** payment is the second most popular having around 19,784 orders. Only, **3,866 orders**, are done by **Vouchers** and **Debit card usage is minimal** having only **1,528 orders**.
- **Nov 2017** has the highest number of orders (**7,713**) and **Dec 2016** has only **1 order**.

By this we can say that customers are mainly rely on credit card payments, this can be because of rewards point or to increase their credit score which can be beneficial for them in the future.

- ii. Find the no. of orders placed on the basis of the payment installments that have been paid.

Query

```
SELECT payment_installments,
COUNT(order_id) AS Num_of_order
FROM `target.payments`
WHERE payment_value > 0
GROUP BY payment_installments
ORDER BY payment_installments;
```

Result

Query results

Job information	Results	Visualise
Row	payment_installm...	Num_of_order
1	0	2
2	1	52537
3	2	12413
4	3	10461
5	4	7098
6	5	5239
7	6	3920
8	7	1626
9	8	4268
10	9	644

Insights

- The highest number of orders (**52,537**) were paid in **1installment**. This indicates that most of the customers are comfortable in paying full amount in one go.

- Small portion prefer 2-6 installments which is nearly about 41,074 orders. These together shows that most of the customers prefer in breaking down the payments, but they still less than the single go customers.
- Very few customers go for the long-term EMI plans like only **133 orders for 12 installments** and there are very few customers who go **installments beyond 12 (less than 30 orders each for 13-24 installments)**.

This shows that most customers prefer to pay the full amount at once. While EMI options are useful for some. The business should focus on the customers who pay in once, since they are in majority.